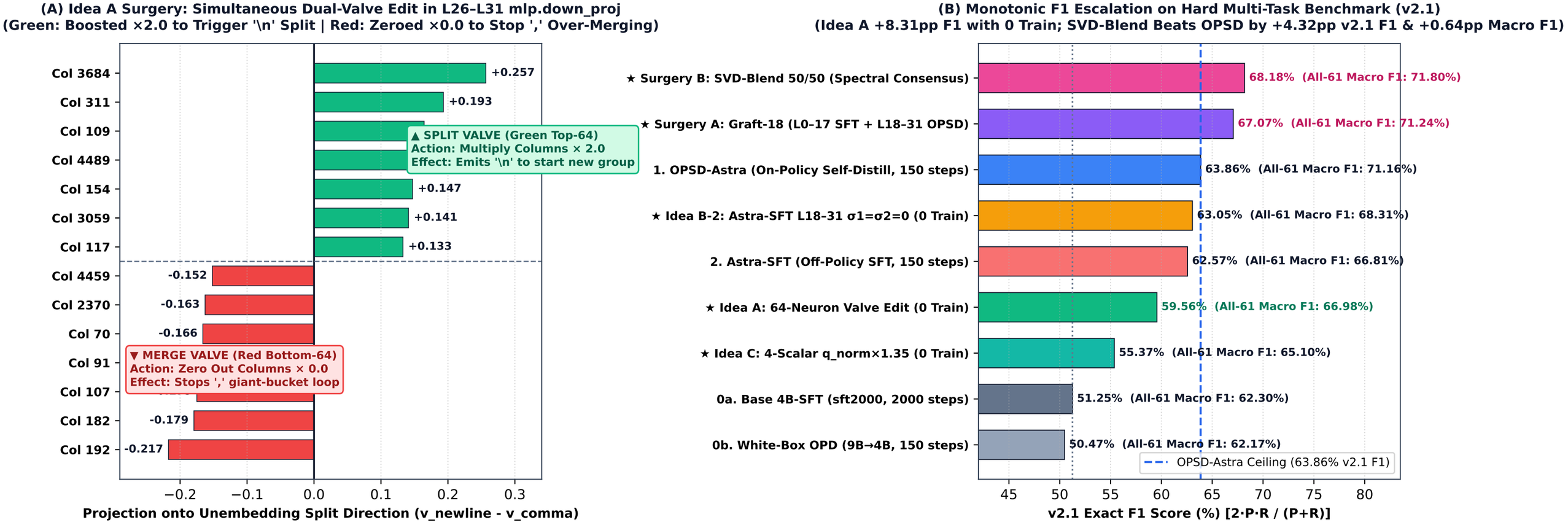
(A) Idea A Surgery: Simultaneous Dual-Valve Edit in L26–L31 mlp.down_proj
(Green: Boosted ×2.0 to Trigger '\n' Split | Red: Zeroed ×0.0 to Stop ',' Over-Merging)
Col 3684
Col 311
Col 109
Col 4489
Col 154
Col 3059
Col 117
Col 4459
Col 2370
Col 70
Col 91
Col 107
Col 182
Col 192
+0.257
+0.193
+0.147
+0.141
+0.133
-0.152
-0.163
-0.166
-0.179
-0.217
▲ SPLIT VALVE (Green Top-64)
Action: Multiply Columns × 2.0
Effect: Emits '\n' to start new group
▼ MERGE VALVE (Red Bottom-64)
Action: Zero Out Columns × 0.0
Effect: Stops ',' giant-bucket loop
−0.2
−0.1
0.0
0.1
0.2
0.3
Projection onto Unembedding Split Direction (v_newline - v_comma)
(B) Monotonic F1 Escalation on Hard Multi-Task Benchmark (v2.1)
(Idea A +8.31pp F1 with 0 Train; SVD-Blend Beats OPSD by +4.32pp v2.1 F1 & +0.64pp Macro F1)
★ Surgery B: SVD-Blend 50/50 (Spectral Consensus)
★ Surgery A: Graft-18 (L0–17 SFT + L18–31 OPSD)
1. OPSD-Astra (On-Policy Self-Distill, 150 steps)
★ Idea B-2: Astra-SFT L18–31 σ1=σ2=0 (0 Train)
2. Astra-SFT (Off-Policy SFT, 150 steps)
★ Idea A: 64-Neuron Valve Edit (0 Train)
★ Idea C: 4-Scalar q_norm×1.35 (0 Train)
0a. Base 4B-SFT (sft2000, 2000 steps)
0b. White-Box OPD (9B→4B, 150 steps)
68.18% (All-61 Macro F1: 71.80%)
67.07% (All-61 Macro F1: 71.24%)
63.86% (All-61 Macro F1: 71.16%)
63.05% (All-61 Macro F1: 68.31%)
62.57% (All-61 Macro F1: 66.81%)
59.56% (All-61 Macro F1: 66.98%)
55.37% (All-61 Macro F1: 65.10%)
51.25% (All-61 Macro F1: 62.30%)
50.47% (All-61 Macro F1: 62.17%)
OPSD-Astra Ceiling (63.86% v2.1 F1)
45
50
55
60
65
70
75
80
v2.1 Exact F1 Score (%) [2·P·R / (P+R)]